



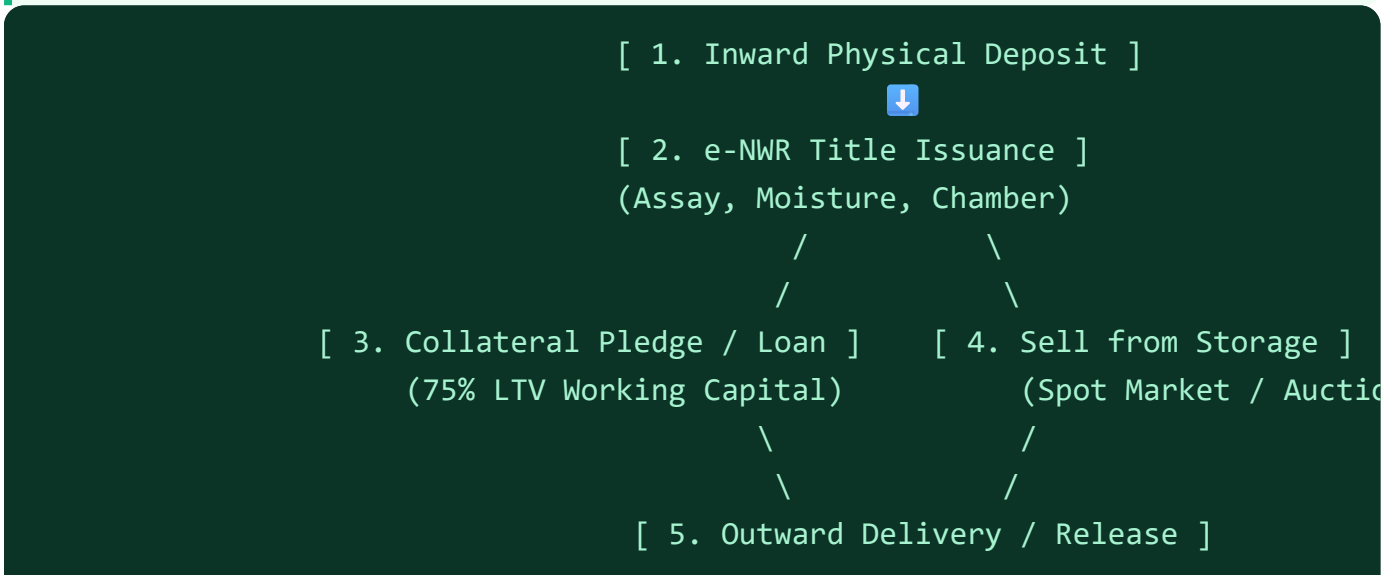
## AgroLnk — Complete Warehouse & e-NWR Workflow Testing Manual

This manual provides an end-to-end testing blueprint for the **WDR-Accredited e-NWR Warehousing, Multi-Chamber Cold Storage, and Commodity Collateral Financing** workflows in AgroLnk.

### 1. Actors & Roles in the Warehouse Ecosystem

| ROLE               | TEST USER EMAIL                    | PASSWORD                      | PRIMARY WORKFLOW ACTIONS                                                                                                                 |
|--------------------|------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Warehouse Operator | <code>warehouse@agrolnk.com</code> | <code>pass123</code> (or any) | View multi-chamber occupancy, audit storage receipts, inspect temperature zones, manage inward/outward gatepasses.                       |
| Verified Farmer    | <code>farmer@agrolnk.com</code>    | <code>farmer123</code>        | Deposit harvested produce, receive digital <b>e-NWR titles</b> , apply for 75% LTV commodity loans, list for sale directly from storage. |
| Financier / NBFC   | <code>financier@agrolnk.com</code> | <code>pass123</code>          | Underwrite working capital against e-NWR receipts, place collateral liens in the <b>Collateral Vault</b> , disburse credit.              |
| Wholesale Buyer    | <code>buyer@agrolnk.com</code>     | <code>buyer123</code>         | Procure warehouse-stored produce with verified assay certificates and immediate delivery/title transfer.                                 |

### 2. Warehouse Life-Cycle Architecture



### 3. Master Step-by-Step Test Scenarios

#### ◆ SCENARIO 1: Farmer Inward Deposit & e-NWR Receipt Creation

**Objective:** Validate that a farmer can choose an accredited warehouse facility, specify produce assay parameters, calculate monthly storage costs, and generate a WDR electronic Negotiable Warehouse Receipt (e-NWR).

Steps for Tester:

Login as Farmer ( farmer@agroInk.com ).

Navigate to Warehouse & e-NWR Vault ( #/farmer-inventory ) or click Inventory in the sidebar.

Click the + Deposit Produce button in the header banner.

In the Deposit to Accredited Warehouse Modal:

Select Warehouse: Salem Agri Cold Storage Hub (WDRA/2024/TN/0892)

Commodity: Tomato

Variety: Hybrid Shivam

Grade: Grade A Premium

Quantity: 1000 kg (1.0 MT)

Estimated Value: ₹42 / kg ( ₹42,000 Total Valuation )

Assigned Chamber: Chamber B4 (Ultra Cold: 2°C - 4°C)

Storage Duration: 60 Days

Verify Cost Estimation Display:

Observe the automated monthly storage fee calculation: 1.0 MT × ₹350/MT = ₹350 / month .

Click Confirm & Generate e-NWR Receipt.

Expected Outcome:

A new electronic receipt (e.g. eNWR-2024-8841 ) is generated instantly.

The receipt is added to the farmer's active inventory with status Stored .

Total stored metric and valuation counters update automatically.

◆ SCENARIO 2: Inspecting e-NWR Title Certificate & Assay Parameters

**Objective:** Confirm that the generated e-NWR contains full legal title ownership, QR/WDRA accreditation, and quality assay data.

Steps for Tester:

On the Farmer Inventory page ( #/farmer-inventory ), locate the newly created receipt card.

Click View e-NWR Certificate.

Verify the Electronic Title Modal:

Title Header: Electronic Negotiable Warehouse Receipt (WDRA Accredited Title)

Receipt ID: Formatted as eNWR-2024-XXXX with an emerald badge.

Chamber & Temperature Specs: Displays exact room temp (e.g., Chamber B4 (2°C - 4°C) ).

Assay Parameters: Moisture %, foreign matter %, and grade classification.

Action CTAs Available:

[ 🛒 Sell from Storage ]

[ 💰 Apply for Working Capital ]

Expected Outcome: High-fidelity, bankable electronic title document viewable on any device.

◆ SCENARIO 3: Warehouse Operator Dashboard & Multi-Chamber Auditing

**Objective:** Verify that the warehouse operator has complete oversight of capacity, chamber temperatures, and stored receipts.

Steps for Tester:

Login as Warehouse Operator ( warehouse@agroInk.com ).

Land on the Warehouse Management & e-NWR Terminal ( #/warehouse-dashboard ).

Verify Metric Cards:

Total Capacity: 5,000 Tonnes

Occupancy %: Dynamically calculated based on active stored tonnes.

**Active e-NWRs:** Count of valid receipts in custody.

**Total Custody Valuation:** Aggregated rupee valuation of all stored commodities.

#### Switch to Chamber Overview Tab:

Verify all 4 temperature zones:

- Chamber A1 (Dry Storage) – Ambient (24°C) (Turmeric, Grains)
- Chamber B2 (Cold Cell) – 4°C - 8°C (Potatoes, Carrots)
- Chamber B4 (Ultra Cold) – 2°C - 4°C (Hybrid Tomatoes, Fruits)
- Chamber C1 (CA Controlled) – 0°C - 2°C (Export Apples, Grapes)

#### Switch to Inventory & Receipts Tab:

Search for the farmer's receipt **eNWR-2024-XXXX**.

Click **Audit e-NWR** to review physical custody specs.

**Expected Outcome:** Real-time visibility into multi-chamber utilization and commodity custody.

### ◆ SCENARIO 4: 75% LTV Working Capital Loan against e-NWR

**Objective:** Validate that farmers can pledge stored commodities to unlock instant working capital without selling early.

#### Steps for Tester:

**Login as Farmer** (**farmer@agrolnk.com**) and go to **e-NWR Vault** (**/farmer-inventory**).

On any stored receipt card, click **Apply for Working Capital** (or open the receipt detail modal and click **Apply for Working Capital**).

In the **e-NWR Collateral Financing Modal**:

Selected Commodity & Valuation: **₹42,000**

**Maximum Loan Facility (75% LTV):** **₹31,500**

Enter Requested Amount: **₹25,000**

Select Tenor: **30 Days** (1.2% monthly interest)

Note: **Pre-harvest liquidity for input fertilizers.**

Click **Submit Working Capital Request**.

**Expected Outcome:** Financing request is routed directly to the Financier / NBFC underwriting queue.

### ◆ SCENARIO 5: NBFC Credit Approval & Collateral Vault Lien

**Objective:** Confirm that the financier reviews the warehouse receipt, places a legal lien on the commodity, and disburses credit.

#### Steps for Tester:

**Login as Financier / NBFC** (**financier@agrolnk.com**).

Go to **Financier Portal** ➔ **Underwriting Desk** (**/financier-underwriting**).

Locate the submitted loan application for the farmer (**₹25,000** backed by **eNWR-2024-XXXX**).

Click **Review & Underwrite**:

Inspect WDRA warehouse accreditation code.

Inspect commodity valuation buffer.

Click **Approve & Disburse Funds**.

**Navigate to Collateral Vault** (**/financier-collateral-vault**):

Verify that the receipt is now listed in the **Collateral Vault** with an active **Lien / Pledge Badge**.

Commodity quantity is locked from unauthorized farmer withdrawal.

**Expected Outcome:** Seamless institutional credit underwriting backed by digital agri-collateral.

◆ SCENARIO 6: Listing for Sale Directly from Warehouse (Zero Transit Latency)

**Objective:** Verify that farmers can list all or part of their stored commodity on the Spot Marketplace or Live Auction directly from the warehouse.

Steps for Tester:

**Login as Farmer** ( farmer@agroInk.com ) and go to **e-NWR Vault** ( #/farmer-inventory ).

On the receipt card, click **Sell from Storage**.

In the **List from Warehouse Modal**:

Choose Sale Type: **Direct Spot Listing** (or **Live Auction**)

Available Stored: 1000 kg

Quantity to List: 500 kg (Partial listing test)

Selling Price: ₹45 / kg

Click **Publish to Marketplace**.

**Verify System Updates:**

The receipt's available quantity decreases to 500 kg , and status reflects partially\_listed .

**Login as Buyer** ( buyer@agroInk.com ) and open **Marketplace** ( #/buyer-marketplace ).

The newly published lot appears on the marketplace marked with a  **Warehouse-Stored (WDRA Certified)** badge, enabling buyers to purchase pre-stored, pre-assayed goods with zero delay.

**Expected Outcome:** Frictionless transition from storage into digital commerce without double handling or freight wastage.

 4. Warehouse & e-NWR QA Verification Checklist

| TEST ITEM                     | VERIFICATION CRITERIA                                                                | STATUS |
|-------------------------------|--------------------------------------------------------------------------------------|--------|
| Accredited Facility Selection | Only WDRA verified hubs appear in the deposit selector.                              | ✓ Pass |
| Storage Fee Formula           | (Weight in Tonnes) × (Rate/Tonne/Month) × (Months) calculated accurately.            | ✓ Pass |
| e-NWR Receipt Numbering       | Clean sequential/timestamped format ( eNWR-YYYY-XXXX ).                              | ✓ Pass |
| Temperature Zone Mapping      | Produce types correctly mapped to designated cold chain chambers.                    | ✓ Pass |
| Partial Quantity Listing      | Stored inventory decrements correctly when a portion is listed on the marketplace.   | ✓ Pass |
| Collateral Lien Protection    | Financier pledge locks receipt from being listed or withdrawn until loan settlement. | ✓ Pass |
| Multi-Role Synchronization    | Updates made by Farmer/Operator/Financier reflect in real time.                      | ✓ Pass |